

# Performance-Focused Design Exercise Packet

Part of ASME IDETC 2025 Workshop:

**Adventures in Tradespace Exploration**

17 August 2025

Group ID: 2050



Annotate, mark-up, and write on this page as needed.

### Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

|        |         |         |         |          |        |       |         |          |         |
|--------|---------|---------|---------|----------|--------|-------|---------|----------|---------|
| 196 IN | 80.2 IN | 79.3 IN | 22.9 IN | 72816 ft | 5.87:1 | 46 IN | 1620 KW | 0.948 IN | 0.98 IN |
|--------|---------|---------|---------|----------|--------|-------|---------|----------|---------|

WIDE, SHORT WHEELBASE, TALL ROOF, VERY LONG SUSPENSION TRAVEL, VERY HIGH TORQUE OUTPUT,  
VERY SHORT GEARING, VERY LARGE TIRES, VERY HIGH ENERGY OUTPUT, VERY HEAVY DUTY  
FRAME, MODERATELY ARMORED

Do you think you could have found a better solution given more time?

NOT WITH THE OPTIONS WE WERE GIVEN IN THE VEHICLE INDEX

What, in this round, stopped you from finding a better solution?

LACK OF TESTING BEING DONE, LIMITED VARIABLES, AND THE LIMITED SUPPLY OF VEHICLES

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**Post-Activity Questions:**

Describe the solution the group has selected – write in the value of the design variables.

|                |               |               |                   |                  |                    |                  |                    |                  |                |
|----------------|---------------|---------------|-------------------|------------------|--------------------|------------------|--------------------|------------------|----------------|
| 110"           | 80.1"         | 66.5"         | 20.8"             | 421 lb-ft        | 2.15:1             | 49"              | 1660 kWh           | .939"            | 1.48"          |
| Very<br>Narrow | Very<br>Short | Very<br>Short | V. Long<br>travel | V. Low<br>output | V. Long<br>gearing | V. Large<br>tire | V. High<br>E. Cap. | V. H.D.<br>Frame | Heavy<br>Armor |

Do you think you could have found a better solution given more time?

No, option 2 was clearly the best as depicted  
in charts

What, in this round, stopped you from finding a better solution?

If anything, the charts since they clearly  
labeled #2 as the best option for all  
requirements

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**Post-Activity Questions:**

Describe the solution the group has selected – write in the value of the design variables

|        |         |         |        |        |       |       |      |       |      |
|--------|---------|---------|--------|--------|-------|-------|------|-------|------|
| 181 in | 81.2 in | 69.2 in | 9.7 in | 655 lb | 2,371 | 32 in | 1079 | 0.877 | 1.99 |
|--------|---------|---------|--------|--------|-------|-------|------|-------|------|

Do you think you could have found a better solution given more time?

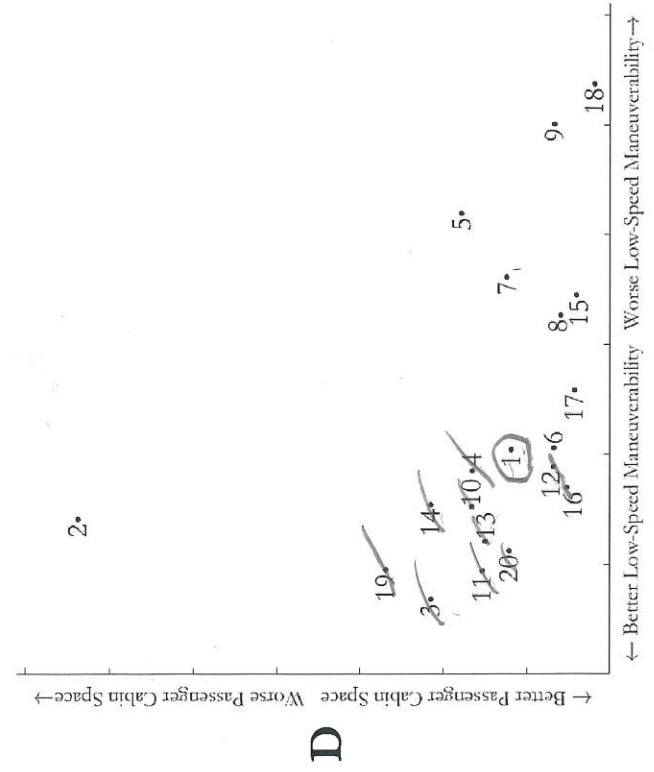
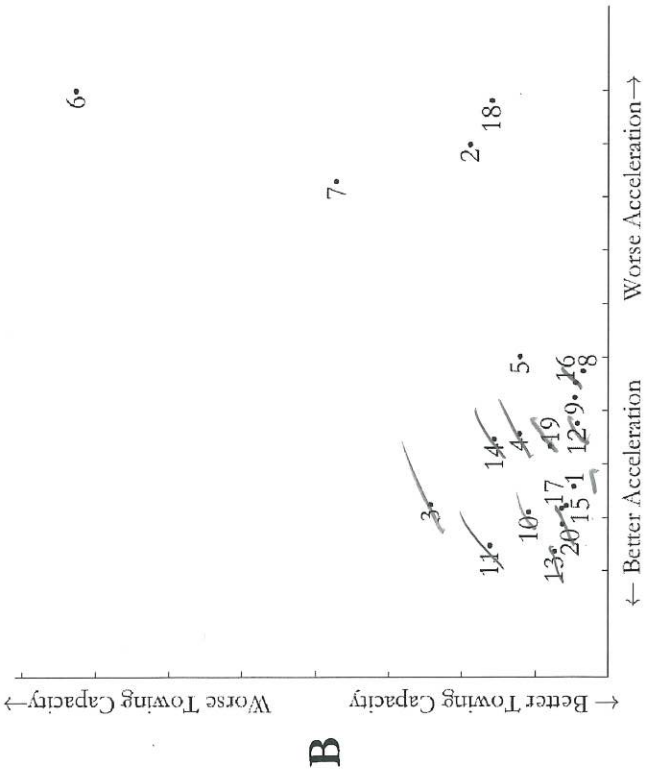
No the clear choice was between 4 and 7

What, in this round, stopped you from finding a better solution?

with out seeing them perform we can't choose the best vehicle.

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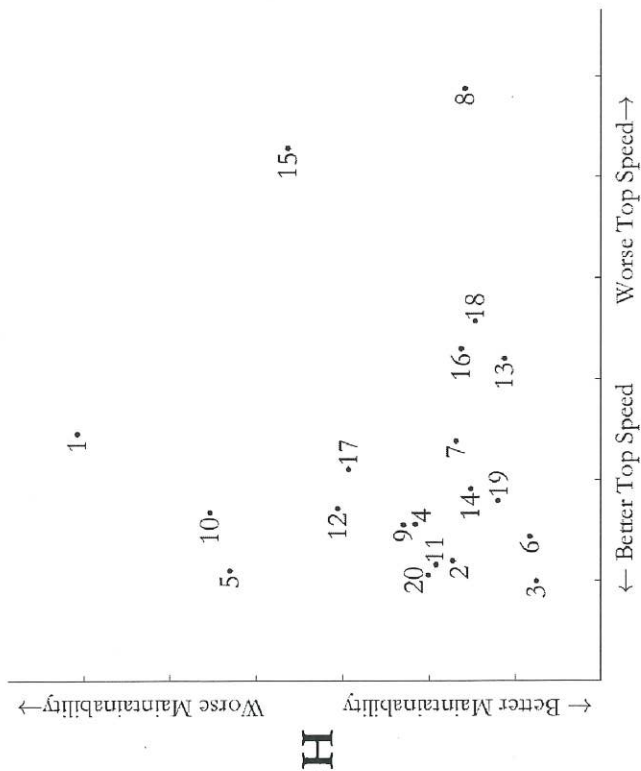


**A**

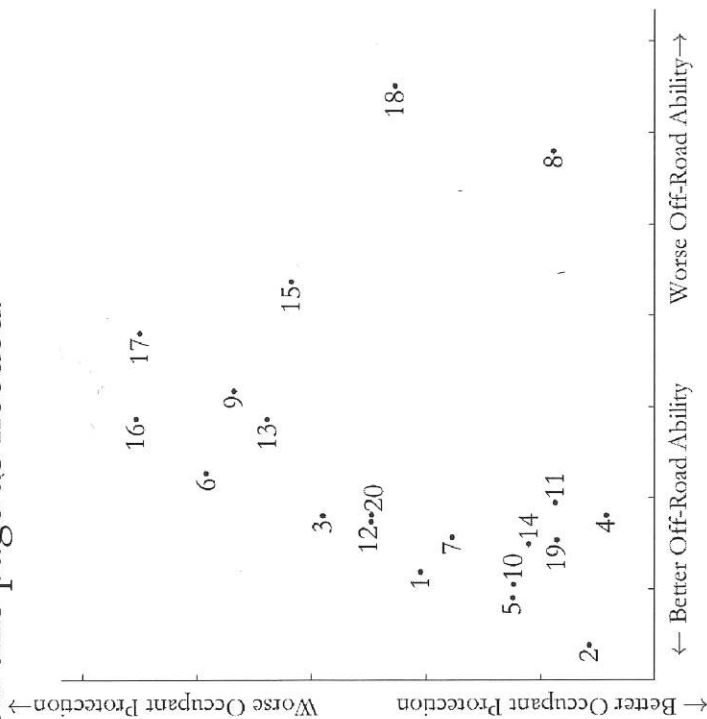
**C**

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**E**

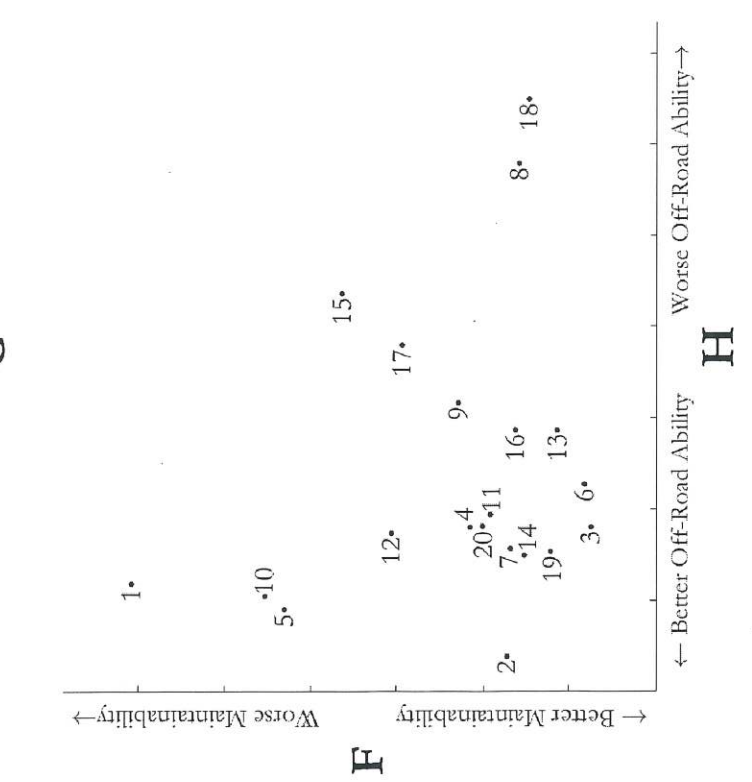
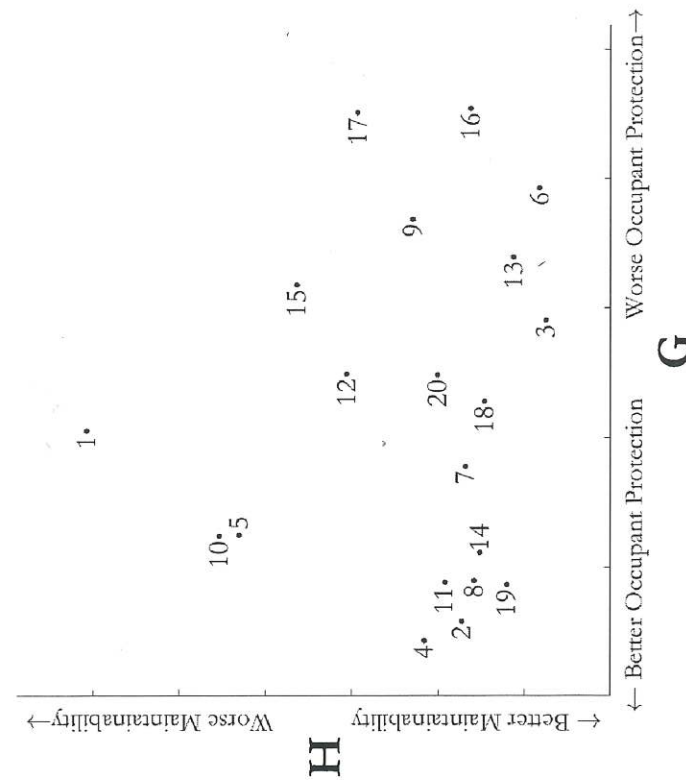
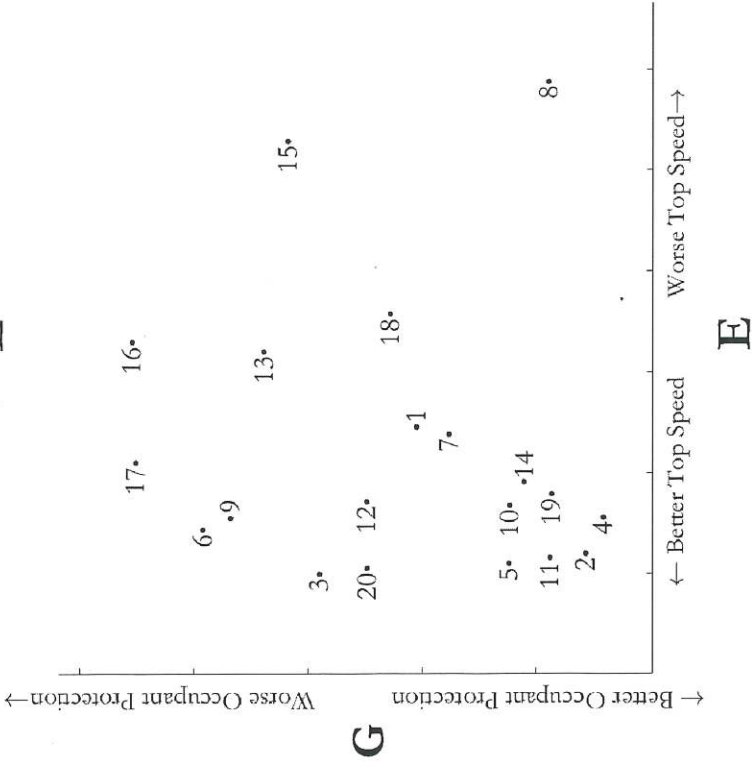
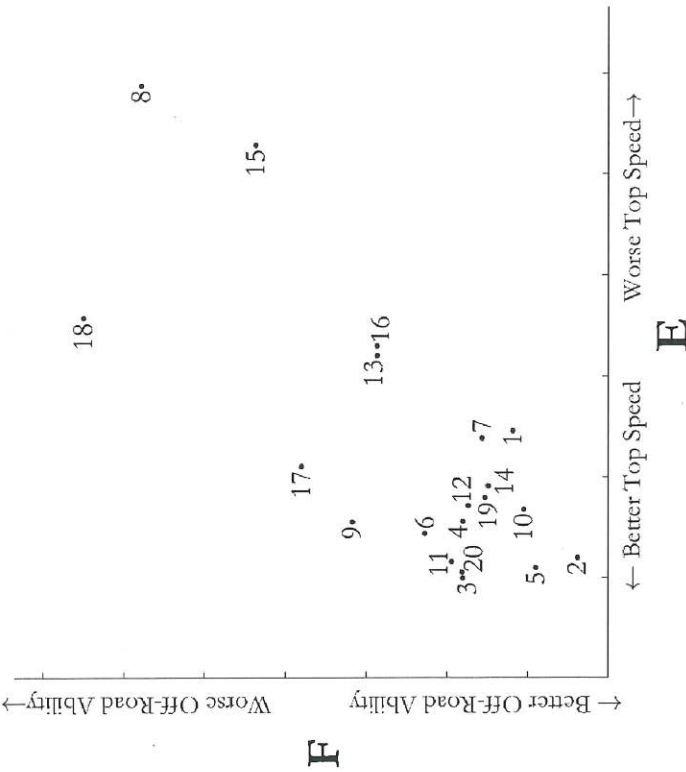


**G**

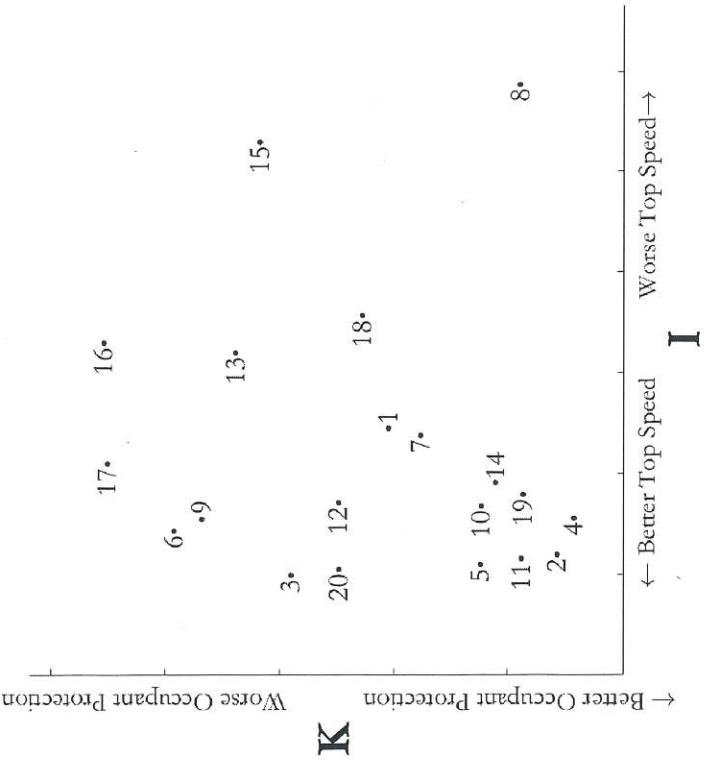
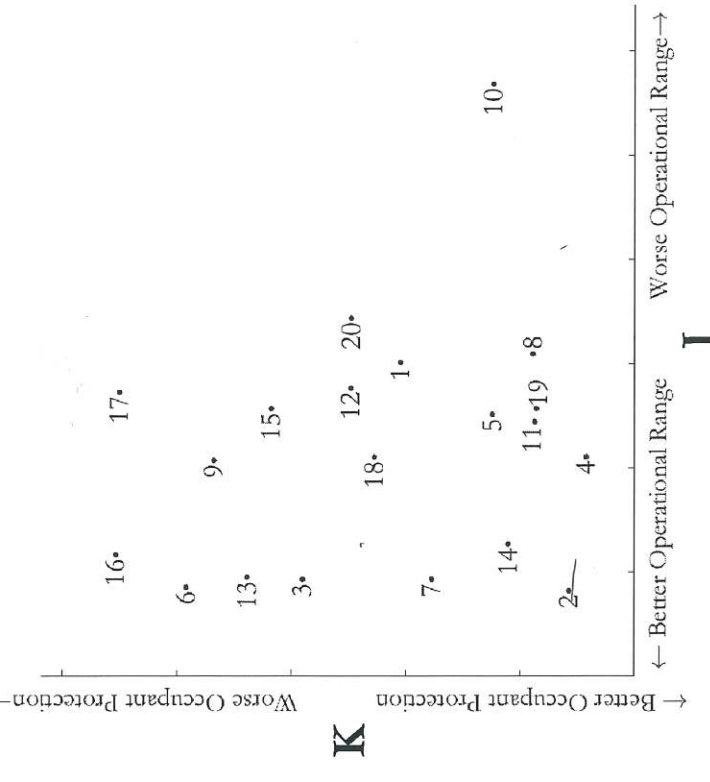
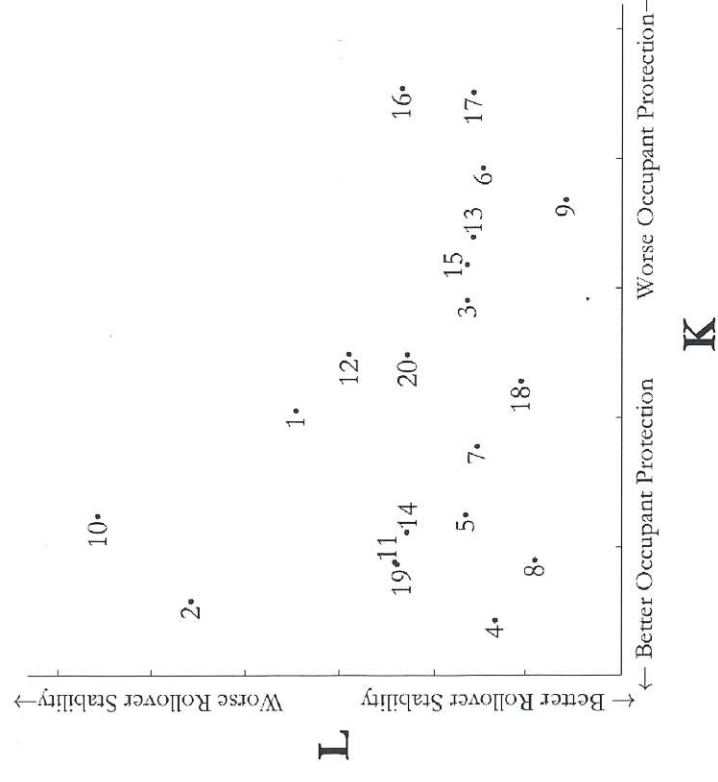
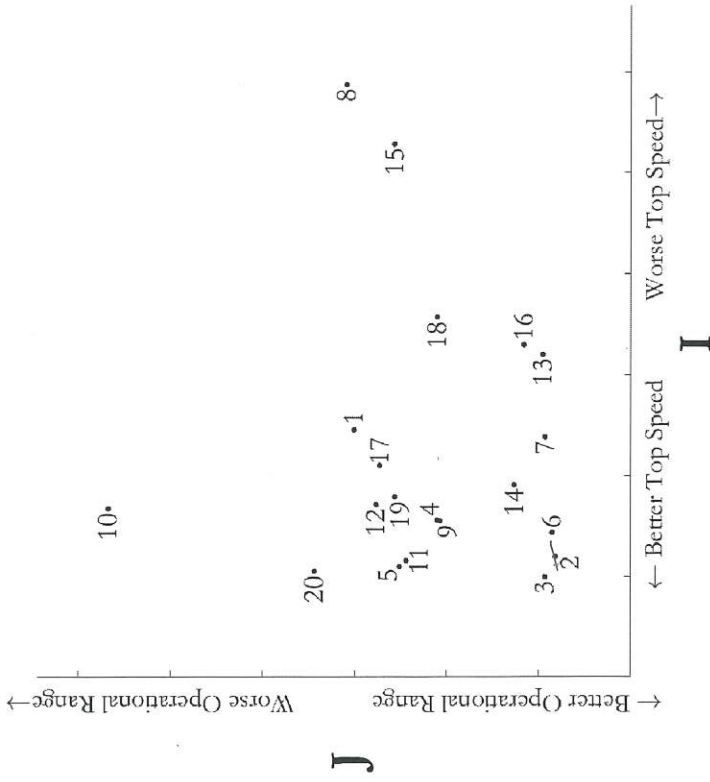
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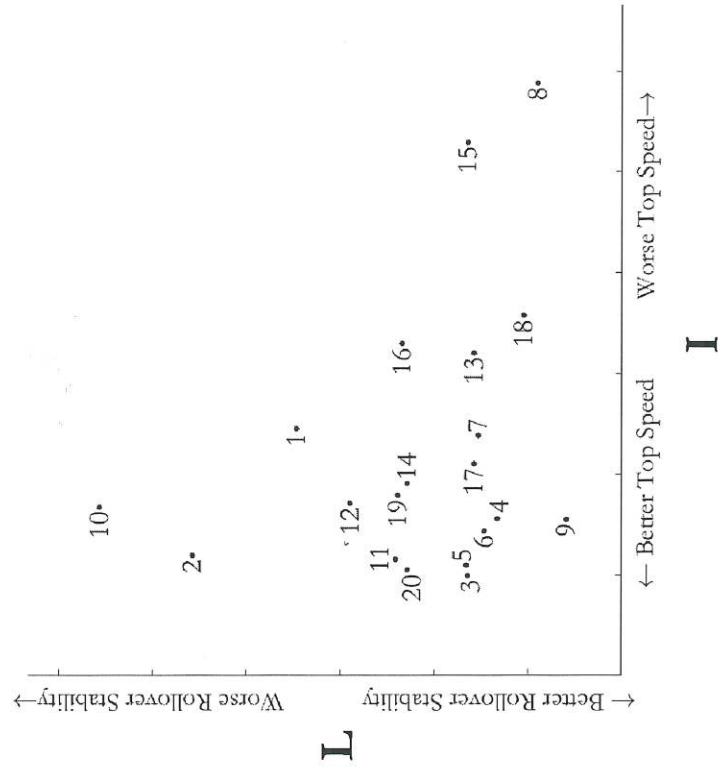
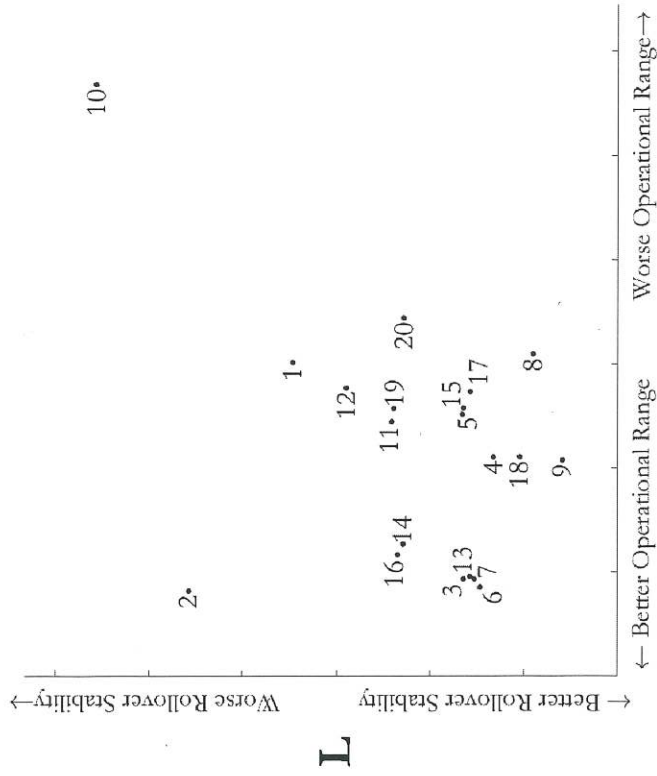


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~~1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18~~  
~~19 20~~

Enter the number ID of the selected design: 4  
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